Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/05/2024
PI Name:	Holden, Kritina Ph.D.		
Project Title:	Communication Delays in Cislunar Space: A Lab Study Examining HSIA Risk Concerns		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058-3607	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
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Task Description:	Astronauts on Lunar Missions will maintain communications with ground support through a communications network that is expected to have communication (comm) delays. While recent analog studies through the Human Research Program have evaluated comm delay effects on performance, much of this work has been completed in context of a Mars mission. As NASA focuses on upcoming Lunar missions, comm delay is still expected to play a part, however, the duration of lunar comm delays is more on the order of seconds rather than minutes. The effects of this level of comm delay are less understood, but the implications may be larger than realized. The purpose of this effort is to help identify ranges of comm delay in context of lunar missions and in the realm of HSIA-related outcomes (e.g., efficiency, effectiveness), where “breaking points” may occur, and to understand responses (e.g., change in performance, workarounds taken) to those delays.		

Rationale for HRP Directed Research:	Results from this assessment will be used to help set the ranges of comm delay further studied in Human Exploration Research Analog (HERA) Campaign 8. Given that HERA Campaign 8 planning will need to be defined in about 15 months, there is not enough time to solicit and complete this pre-cursor effort. As a result, HFBP is initiating this task as a directed task.
Research Impact/Earth Benefits:	The team has developed a technology setup to simulate communication delay of audio and video. This capability can be used in future communication delay studies, and the setup is flexible enough to simulate a range of different delays. The interactive display prototypes are available for use in other studies, and in fact, are already being leveraged by another HFBP directed research project needing a realistic spacecraft experimental task. Results will inform the need for countermeasures for effective and efficient task performance under lunar communication delay.
Task Progress:	Research articles on communication delay were reviewed and discussed by the team. Very few studies have been done using delays anticipated for upcoming lunar missions. Based on this literature and discussions with NASA experts on the expected lunar communication systems, the team decided to test using delays of 0,4,6,8,10, and 12 seconds. A study was designed to test the performance and problem-solving of a crewmember/flight controller team collaborating to perform realistic spaceflight tasks under different levels of communication delay. Seven interactive prototypes were developed for the study, using Axure prototyping software. The 7 prototypes provided 6 unique task experiences for each of the 6 communication delay conditions (0s, 4s, 6s, 8s, 10s, 12s), plus one slightly different prototype for the flight controller participant. The experimental prototypes were developed to be high-fidelity replicas of spacecraft displays. The tasks and associated procedures were also high fidelity and consisted of realistic nominal and off-nominal computer-based display tasks (i.e., fault resolution, procedure issue resolution). In addition to video recording each participant's displays, the prototypes provided automated data collection capability (i.e., click recording and timing). In addition to the prototypes, Qualtrics surveys were developed to collect demographic and background information, as well as to collect ratings on scales, such as workload, situation awareness, and usability. The surveys also included questions specific to the topic of communication delay. Experimental sessions have just begun. Each participant will complete 6 different sessions, separated by approximately 2 weeks; thus, subject running will continue over several months.
Bibliography Type:	Description: (Last Updated: 10/29/2023)